# Arcs complete outside a conic: a prescribed-hole defect identity and matching-design rigidity

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**Abstract**

For a nonsingular conic $\mathcal{C} \subseteq \mathrm{PG}(2,q)$, let $\rho_{\mathcal{C}}(q)$ be the least size of an arc disjoint from $\mathcal{C}$ whose secants cover every remaining point outside $\mathcal{C}$. We prove

$$\rho_{\mathcal{C}}(q) \geq \sqrt{2q} + \frac{3}{2} - \frac{8}{\sqrt{2q}}.$$

We determine the possible zero-defect orders and classify zero defect in characteristic two: odd $k$ gives an oval whose nucleus lies on $\mathcal{C}$, while even $k$ forces $k = q+2$. A kernel-checked finite classification proves $\rho_{\mathcal{C}}(16) = 9$. Separate exhaustive classifications, with kernel-checked attaining witnesses, give $\rho_{\mathcal{C}}(13) = 8$, $\rho_{\mathcal{C}}(17) = 9$, and $\rho_{\mathcal{C}}(19) = 10$.

The mechanism is a theorem for every finite projective plane and every prescribed hole set $\mathcal{H}$: the first two secant moments combine into an exact identity with a pointwise nonnegative remainder. Its vanishing turns the canonical secant-concurrency decomposition of $KG(k,2)$ into a simple MATCH$(k, \lfloor k/2 \rfloor, 1)$ design; in a Desarguesian plane this design has a rank-three projective realization. Its total gives an explicit deletion bound away from equality. The defect identity, equality criterion, and deletion stability are independently formalized in Lean.

## 1 Introduction

An arc in a projective plane is a set of points no three of which are collinear. An arc is complete when it cannot be enlarged while remaining an arc; equivalently, every point outside it lies on a secant determined by two of its points. Complete arcs and the closely related problem of finding small saturating sets have a substantial literature; see, for example, Ball [4], Kim–Vu [11], and Nagy [14].

More generally, one may prescribe a set of points that are allowed to remain uncovered. The exact defect identity for this problem, valid in every finite projective plane and for an arbitrary prescribed set, is the main result of the paper. Its principal application here takes the prescribed set to be a nonsingular conic $\mathcal{C} \subset \mathrm{PG}(2,q)$.

**Definition 1.1.** An arc $A \subseteq \mathrm{PG}(2,q) \setminus \mathcal{C}$ is *$\mathcal{C}$-complete* if every point of

$$\mathrm{PG}(2,q) \setminus (A \cup \mathcal{C})$$

lies on a secant of $A$. Define

$$\rho_{\mathcal{C}}(q) = \min\{|A| : A \text{ is a } \mathcal{C}\text{-complete arc}\}.$$

More generally, if $\mathcal{H}$ is any set disjoint from an arc $A$, write

$$\mathcal{U}_{\mathcal{H}}(A) = \mathrm{PG}(2,q) \setminus \left( A \cup \mathcal{H} \cup \bigcup_{\{a,b\} \in \binom{A}{2}} ab \right).$$

We abbreviate $U(A) = \mathcal{U}_\varnothing(A)$ for the ordinary-uncovered locus and retain $\mathcal{U}_\mathcal{C}(A)$ for the conic-relative locus. Thus $A$ is $\mathcal{C}$-complete exactly when $\mathcal{U}_\mathcal{C}(A) = \varnothing$.

In a Desarguesian plane, projective representatives of the points of $A$ form a parity-check matrix of a $[k, k-3, 4]_q$ MDS code. If $U(A) \neq \varnothing$, then $U(A)$ is exactly its projective deep-hole syndrome locus: points of $A$, points on secants, and points of $U(A)$ have coset weight one, two, and three, respectively.

The minimum is attained because the finite point set $\mathrm{PG}(2,q) \setminus \mathcal{C}$ contains an inclusion-maximal arc $A$. For every point $x \notin A \cup \mathcal{C}$, maximality forces $A \cup \{x\}$ to contain three collinear points, so $x$ lies on a secant of $A$. Thus $A$ is $\mathcal{C}$-complete. All nonsingular conics in $\mathrm{PG}(2,q)$ are projectively equivalent, so the numerical value of $\rho_\mathcal{C}(q)$ does not depend on the chosen conic.

For a point $x$ outside a $k$-arc $A$, let $r(x)$ be the number of secants of $A$ through $x$, and put $m = \lfloor k/2 \rfloor$ and $N = \binom{k}{2}$. If a set $\mathcal{H}$ disjoint from $A$ is allowed to remain uncovered, let $\mathcal{X}_\mathcal{H}(A)$ be the covered points outside $A \cup \mathcal{H}$ and let $I_\mathcal{H}(A) = \sum_{y \in \mathcal{H}} r(y)$.

**Theorem 1.2** (Exact prescribed-hole defect)**.** *For every arc $A$ and prescribed hole $\mathcal{H}$, define*

$$\Delta_\mathcal{H}(A) = N(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_\mathcal{H}(A)}{m} - |\mathcal{X}_\mathcal{H}(A)|.$$

*Then*

$$m\Delta_\mathcal{H}(A) = \sum_{x \in \mathcal{X}_\mathcal{H}(A)} (r(x)-1)(m-r(x)) + \sum_{y \in \mathcal{H}} r(y)(m-r(y)).$$

**Corollary 1.3** (Equality rigidity and deletion stability)**.** *The concurrency cliques of the secants canonically decompose $KG(k,2)$. For $k \geq 4$, if $\Delta_\mathcal{H}(A) = 0$, these cliques are maximum matchings and form a simple* $\mathrm{MATCH}(k, \lfloor k/2 \rfloor, 1)$ *design with a star–matching incidence realization in the dual plane. When the plane is Desarguesian, this is a rank-three projective realization over its underlying field. For $k \geq 4$, the defect controls both the edge count and the vertex-cover number of the graph formed by the nonmaximum concurrency cliques; after deleting at most $m\Delta_\mathcal{H}(A)$ secants, every surviving Kneser edge lies in a maximum concurrency clique.*

**Theorem 1.4** ($q+1$-hole lower bound and exact small orders)**.** *If*

$$L_2(q) = \min\left\{k \geq 3 : q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{\lfloor k/2 \rfloor}\binom{k}{4}\right\},$$

*then every $k$-arc complete outside a prescribed set of $q+1$ points satisfies $k \geq L_2(q)$. In particular, every nonsingular conic satisfies*

$$\rho_\mathcal{C}(q) \geq L_2(q) \geq \sqrt{2q} + \frac{3}{2} - \frac{8}{\sqrt{2q}}.$$

*Moreover,*

$$\rho_\mathcal{C}(13) = 8, \qquad \rho_\mathcal{C}(16) = 9, \qquad \rho_\mathcal{C}(17) = 9, \qquad \rho_\mathcal{C}(19) = 10.$$

The order-16 lower exclusion in this theorem is an exhaustive Lean-kernel-checked certificate. At orders $13, 17, 19$, Lean checks the attaining witnesses and upper bounds, while the exhaustive lower exclusions are trusted classifier executions; the evidence map in Appendix F records the distinction.

The identity carries more information than the resulting scalar bound. The algebra is short; the point is the linear combination that factors pointwise into nonnegative terms.

Its summands record exactly where a secant-index distribution falls short of the extremal pattern. Their vanishing produces the matching design used for equality rigidity, while their total gives stability. For conic holes, the equality theory forces each parity of $k$ into three possible plane orders. In characteristic two the odd branch is exactly the oval–nucleus configuration, while the even branch leaves only the hyperoval scale, by a tangent-involution congruence on the maximum-index points of the prescribed conic.

The exact small-order assertions in Theorem 1.4 have separate finite inputs. At order 16, a checked augmentation certificate enumerates the eight-arc leaves, and checked quadratic-evaluation records reject every leaf. Exhaustive projective classifications reject size 7 at order 13, size 8 at order 17, and sizes $8, 9$ at order 19; the attaining witnesses are checked independently in Lean. Appendix F states the two trust boundaries separately.

The surrounding notions differ at the defining containment. Saturating sets need not be arcs, while almost-complete conic subsets select points on the conic [6]. A line hole recovers affine completeness, including the related hyperfocused constructions [15, 8, 12]; see Corollary 3.11. Complete exterior sets point in the opposite direction: their joins miss the conic [7, p. 143]. Thus complete exteriority gives $\mathcal{C}(\mathbb{F}_q) \subseteq U(A)$, whereas $\mathcal{C}$-completeness is equivalent to $U(A) \subseteq \mathcal{C}(\mathbb{F}_q)$.

The point-index equations themselves are classical; compare Hirschfeld [10, Chapter 9], Ball [4], and the recent explicit complete-arc bound of Alabdullah–Hirschfeld [1]. The new step is the exact local remainder after splitting those equations over a prescribed hole and its complement.

Localization of an uncovered locus on a proper subgeometry is also known for curve-derived arcs with larger line intersections. Korchmáros, Nagy, and Szőnyi [13, Theorem 7.5] prove that, for odd $q$ and odd $r \geq 5$, the points left uncovered by the $(q+1)$-secants of their rational BKS $(k, q+1)$-arc in $\mathrm{PG}(2, q^r)$ are exactly the points of the proper subplane $\mathrm{PG}(2, q)$ not belonging to the arc. Thus localization of holes in a prescribed subplane is prior art. Their theorem concerns a particular curve-derived $(k, q+1)$-arc; the contribution here is instead the exact defect identity for an arbitrary prescribed hole set, together with its equality and stability theory and its specialization to a prescribed conic for ordinary arcs.

The equality structure belongs to the established theory of matching designs. Alspach and Heinrich [3] call a family of $s$-edge matchings in $K_n$ a $\mathrm{MATCH}(n, s, \lambda)$ design when every pair of independent edges occurs in exactly $\lambda$ members. Their framework generalizes hyperfactorizations, which were introduced in connection with higher block designs. Here the prescribed-hole equality condition selects the much narrower class whose matching blocks arise as secant concurrences in one projective plane. Our contribution at this interface is the geometric implication: vanishing of the prescribed-hole defect canonically produces a simple $\mathrm{MATCH}(k, \lfloor k/2 \rfloor, 1)$ design, represented by concurrence points of secants in one projective plane. The local remainder controls both the number of graph edges that fail to lie in such maximum-matching cliques and the number of secants needed to meet all those edges. After projective duality, Corollary 3.7 upgrades the matching design to an incidence realization of the full star–matching pairwise-balanced design. In a Desarguesian plane it is a rank-three realization over the underlying field, providing the bridge to the field and characteristic restrictions recorded in Appendix A.

Sections 2–4 give the main route: exact accounting, equality rigidity, and the conic lower bound. The appendices contain the finite proof of the $q = 16$ result together with the secondary realization classification, upper-bound transfer, characteristic-two incidence lemmas, reconstruction theorem, and verification contracts.

## 2 The classical secant equations

Throughout the remainder of the paper, $q \geq 3$ and $k \geq 3$. Let $\Pi$ be a projective plane of order $q$, and let $A$ be a $k$-arc in $\Pi$. Write $r_A(x)$ for the number of secants of $A$ through $x \in \Pi \setminus A$. We usually write $r(x)$. Put

$$N = \binom{k}{2}, \qquad m = \left\lfloor \frac{k}{2} \right\rfloor.$$

**Lemma 2.1.** *For every $x \notin A$, one has $r(x) \leq m$.*

*Proof.* Distinct secants through $x$ cannot share an endpoint in $A$. Indeed, two lines through the same pair $x, a$ coincide. Hence the secants through $x$ use pairwise disjoint pairs of points of $A$, of which there are at most $\lfloor k/2 \rfloor$. □

**Proposition 2.2** (Classical first and second index equations)**.** *For every $k$-arc $A$ in a projective plane of order $q$,*

$$\sum_{x \notin A} r(x) = \binom{k}{2}(q-1), \tag{1}$$

$$\sum_{x \notin A} \binom{r(x)}{2} = 3\binom{k}{4}. \tag{2}$$

*Proof.* There are $\binom{k}{2}$ secants. Each contains $q+1$ points, exactly two of them in $A$, and therefore contributes $q-1$ incidences with points outside $A$. This proves (1).

For (2), count unordered pairs of distinct secants through a common external point. By Lemma 2.1, their four endpoints in $A$ are distinct. Conversely, a four-subset of $A$ has three partitions into two unordered pairs. Each partition gives two secants, which meet in a unique point because the plane is projective. Their intersection is outside $A$: otherwise one of the two secants would contain three points of the arc. Thus every four-subset contributes exactly three. □

Only the projective-plane axioms and the common line size $q+1$ enter these proofs. In particular, the second equation need not hold in a general finite linear space where two lines may be disjoint.

## 3 An exact defect identity for prescribed holes

The next theorem is stated for an arbitrary exceptional point set. The conic enters only in later specializations.

Let $\mathcal{H} \subseteq \Pi \setminus A$ be any set of points allowed to remain uncovered. The notation used in the identity is summarized by

| | |
|---|---|
| $V_{\mathcal{H}}(A) = \Pi \setminus (A \cup \mathcal{H})$ | required locus |
| $\mathcal{X}_{\mathcal{H}}(A) = \{x \in V_{\mathcal{H}}(A) : r(x) > 0\}$ | covered required points |
| $\mathcal{U}_{\mathcal{H}}(A) = V_{\mathcal{H}}(A) \setminus \mathcal{X}_{\mathcal{H}}(A)$ | uncovered required points |
| $I_{\mathcal{H}}(A) = \sum_{y \in \mathcal{H}} r(y)$ | secant incidence spent on the holes. |

The third line agrees with the uncovered-locus notation introduced in Definition 1.1; in particular, $U = \mathcal{U}_{\varnothing}$ and the conic-relative locus is $\mathcal{U}_{\mathcal{C}}$.

Splitting Proposition 2.2 over $V_{\mathcal{H}}(A)$ and $\mathcal{H}$ gives

$$\sum_{x\in V_{\mathcal{H}}(A)} r(x) = N(q-1) - I_{\mathcal{H}}(A), \tag{3}$$

$$\sum_{x\in V_{\mathcal{H}}(A)} \binom{r(x)}{2} + \sum_{y\in\mathcal{H}} \binom{r(y)}{2} = 3\binom{k}{4}. \tag{4}$$

**Theorem 3.1** (Prescribed-hole defect identity)**.** *Let $A$ be a $k$-arc in a projective plane of order $q$, let $\mathcal{H} \subseteq \Pi \setminus A$, and put $m = \lfloor k/2 \rfloor$. Define*

$$\Delta_{\mathcal{H}}(A) = N(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m} - |\mathcal{X}_{\mathcal{H}}(A)|.$$

*It is the exact coverage capacity left after the universal second-moment and hole-incidence losses; the identity below resolves that capacity into local nonextremal-index contributions. Then*

$$\boxed{m\Delta_{\mathcal{H}}(A) = \sum_{x\in\mathcal{X}_{\mathcal{H}}(A)} (r(x)-1)(m-r(x)) + \sum_{y\in\mathcal{H}} r(y)(m-r(y)).} \tag{5}$$

*In particular, $\Delta_{\mathcal{H}}(A) \geq 0$.*

*Proof.* Only points of $\mathcal{X}_{\mathcal{H}}(A)$ contribute to the sum in (3). Multiplying the definition of $\Delta_{\mathcal{H}}(A)$ by $m$ and using (3) gives

$$m\Delta_{\mathcal{H}}(A) = m\sum_{x\in\mathcal{X}_{\mathcal{H}}(A)} r(x) + (m-1)I_{\mathcal{H}}(A) - m|\mathcal{X}_{\mathcal{H}}(A)| - 6\binom{k}{4}.$$

By (4),

$$6\binom{k}{4} = 2\sum_{x\in\mathcal{X}_{\mathcal{H}}(A)} \binom{r(x)}{2} + 2\sum_{y\in\mathcal{H}} \binom{r(y)}{2}.$$

Substitution first separates the required-point and hole terms:

$$\begin{aligned} m\Delta_{\mathcal{H}}(A) = &\sum_{x\in\mathcal{X}_{\mathcal{H}}(A)} \left( m(r(x)-1) - 2\binom{r(x)}{2} \right) \\ &+ \sum_{y\in\mathcal{H}} \left( (m-1)r(y) - 2\binom{r(y)}{2} \right). \end{aligned}$$

The pointwise identities

$$\begin{aligned} m(r-1) - 2\binom{r}{2} &= (r-1)(m-r), \\ (m-1)r - 2\binom{r}{2} &= r(m-r), \end{aligned}$$

now give (5). Every term is nonnegative by Lemma 2.1. $\square$

For required points, the identity records the total slack in the pointwise quadratic inequality

$$\mathbf{1}_{\{r>0\}} \leq r - \frac{2}{m}\binom{r}{2} \qquad (0 \leq r \leq m),$$

whose positive-index equality cases are exactly $r = 1$ and $r = m$. Thus the factored remainder is a sharp two-moment certificate, rather than only a rearrangement of the global counts.

**Corollary 3.2** (Coverage and uncovered-locus bounds)**.** *Under the hypotheses of Theorem 3.1,*

$$|\mathcal{X}_{\mathcal{H}}(A)| \le N(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m}, \tag{6}$$

*and*

$$|\mathcal{U}_{\mathcal{H}}(A)| \ge |V_{\mathcal{H}}(A)| - N(q-1) + \frac{6}{m}\binom{k}{4} + \frac{I_{\mathcal{H}}(A)}{m}. \tag{7}$$

*Equality holds in either inequality if and only if*

$$r(x) \in \{1, m\} \quad (x \in \mathcal{X}_{\mathcal{H}}(A)), \qquad r(y) \in \{0, m\} \quad (y \in \mathcal{H}).$$

*Proof.* The two inequalities are respectively $\Delta_{\mathcal{H}}(A) \ge 0$ and its rearrangement using $|\mathcal{U}_{\mathcal{H}}(A)| = |V_{\mathcal{H}}(A)| - |\mathcal{X}_{\mathcal{H}}(A)|$. The equality statement follows from the vanishing of every nonnegative term in (5). ☐

For later use, we record the complete-outside consequence before imposing any geometry on the holes.

**Corollary 3.3** (Arbitrary prescribed holes)**.** *Let $|\mathcal{H}| = h$. If $A$ is a $k$-arc disjoint from $\mathcal{H}$ and complete outside $\mathcal{H}$, then*

$$q^2 + q + 1 - k - h \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m}. \tag{8}$$

*Proof.* Completeness gives $|\mathcal{X}_{\mathcal{H}}(A)| = |V_{\mathcal{H}}(A)| = q^2 + q + 1 - k - h$. Substitute this equality into (6). ☐

**Corollary 3.4** (Any $q+1$ prescribed holes)**.** *Let $|\mathcal{H}| = q + 1$. If a $k$-arc $A$ is disjoint from $\mathcal{H}$ and complete outside $\mathcal{H}$, then*

$$q^2 - k \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4}.$$

*Consequently $k \ge L_2(q)$. No geometric property of $\mathcal{H}$ is used.*

*Proof.* Corollary 3.3 gives the stronger inequality with the additional nonpositive term $-I_{\mathcal{H}}(A)/m$. Drop that term. ☐

The exact remainder also quantifies near equality.

**Corollary 3.5** (Quantitative stability)**.** *Assume $m \ge 3$ and put*

$$M = \{x \in \mathcal{X}_{\mathcal{H}}(A) : 2 \le r(x) \le m-1\},$$
$$J = \{y \in \mathcal{H} : 1 \le r(y) \le m-1\}.$$

*Then*

$$(m-2)|M| + (m-1)|J| \le m\Delta_{\mathcal{H}}(A). \tag{9}$$

*Proof.* For $2 \le r \le m-1$, $(r-1)(m-r) \ge m-2$. For $1 \le r \le m-1$, $r(m-r) \ge m-1$. Apply these estimates to (5). ☐

In particular, the defect has a discrete gap:

$$m\Delta_{\mathcal{H}}(A) = 0 \quad \text{or} \quad m\Delta_{\mathcal{H}}(A) \ge m-2.$$

Thus small defect forces almost every multiply covered required point to have the maximum possible multiplicity $m$, while almost every hole point met by a secant also has multiplicity $m$. This is the precise stability content behind the equality heuristic.

### 3.1 Concurrency designs at equality

The equality criterion has a global incidence interpretation. Label the vertices of $K_k$ by the points of $A$, so that its edges label the secants of $A$. Two such edges are adjacent in the Kneser graph $KG(k,2)$ exactly when the corresponding secants have disjoint endpoint pairs.

**Theorem 3.6** (Concurrency decomposition and equality rigidity)**.** *Suppose $k \geq 4$. For each $x \notin A$ with $r(x) \geq 2$, the secants through $x$ form an $r(x)$-edge matching $M_x$ in $K_k$, and*

$$E\big(KG(k,2)\big) = \dot{\bigcup_{x\notin A,\ r(x)\geq 2}} E(K_{M_x}).$$

*Here $K_{M_x}$ is the clique whose vertices are the edges of $M_x$.*

*If $\Delta_{\mathcal{H}}(A) = 0$, every $M_x$ in this decomposition has $m = \lfloor k/2 \rfloor$ edges. The family is simple: a repeated matching would repeat each of its pairs of edges, contradicting $\lambda = 1$. Thus $(M_x)_{r(x)=m}$ is a simple* MATCH$(k,m,1)$ *design. If $Z = \{x \notin A : r(x) = m\}$, then*

$$|Z| = \frac{3\binom{k}{4}}{\binom{m}{2}} = \begin{cases} (k-1)(k-3), & k \text{ even}, \\ k(k-2), & k \text{ odd}, \end{cases}$$

*and every secant of $A$ contains $k-3$ points of $Z$ when $k$ is even and $k-2$ points of $Z$ when $k$ is odd.*

*Proof.* Secants through a point outside an arc have pairwise disjoint endpoint pairs, so they form a matching. Conversely, two secants with disjoint endpoint pairs meet in a unique point outside $A$. Their edge in $KG(k,2)$ therefore belongs to exactly one concurrence clique, proving the decomposition.

If the defect vanishes, Theorem 3.1 gives $r(x) \in \{1, m\}$ off $\mathcal{H}$ and $r(x) \in \{0, m\}$ on $\mathcal{H}$. Every concurrence point has index at least two, so all the cliques above have order $m$. The second index equation now gives $|Z|\binom{m}{2} = 3\binom{k}{4}$.

Fix a secant. It is disjoint from $\binom{k-2}{2}$ other secants, and each point of $Z$ on it accounts for $m-1$ of them. Hence it contains $\binom{k-2}{2}/(m-1)$ points of $Z$, which has the two displayed values. ☐

**Corollary 3.7** (Dual star–matching incidence realization)**.** *If $\Delta_{\mathcal{H}}(A) = 0$, then after any projective duality the $\binom{k}{2}$ secants become points carrying an incidence realization of the star–matching pairwise-balanced design on $E(K_k)$. Its lines are the $k$ star lines of size $k-1$, one for each point of $A$, and the $|Z|$ matching lines of size $m = \lfloor k/2 \rfloor$. If the original plane is Desarguesian over $K$, this is a rank-three projective realization over $K$.*

*Proof.* The dual secant-points for two edges of $K_k$ lie on a star line when the edges share a vertex. When the edges are disjoint, their secants meet at a unique point outside $A$; Theorem 3.6 places them on a unique matching line, of size $m$ at zero defect. ☐

The same decomposition gives two distances from equality. Let $G_{\text{bad}}$ be the spanning subgraph of $KG(k,2)$ whose edges lie in concurrence cliques of order $2 \leq r(x) \leq m-1$, and write $\tau(G_{\text{bad}})$ for its vertex-cover number. Thus its vertices are the secants of $A$, and a vertex cover is a set of secants meeting every Kneser edge in a nonmaximum concurrence clique.

**Corollary 3.8** (Edge and vertex stability)**.** *For $k \geq 4$,*

$$|E(G_{\text{bad}})| \leq \frac{m(m-1)}{2}\,\Delta_{\mathcal{H}}(A), \qquad \tau(G_{\text{bad}}) \leq m\Delta_{\mathcal{H}}(A).$$

*The second bound says equivalently that there is a set $D \subseteq \binom{A}{2}$, with $|D| \le m\Delta_{\mathcal{H}}(A)$, such that every two secants outside $D$ with disjoint endpoint pairs meet at a point of secant index $m$.*

*Proof.* At an off-hole centre of index $r$,

$$\binom{r}{2} \le \frac{m-1}{2}(r-1)(m-r).$$

For a hole centre whose clique contributes to $G_{\mathrm{bad}}$, its defect weight $r(m-r)$ is at least the off-hole weight $(r-1)(m-r)$. Summing over the edge-disjoint concurrence cliques and applying (5) proves the edge bound.

At each concurrence centre $x$ with $2 \le r(x) \le m-1$, retain one secant through $x$ and put the other $r(x)-1$ secants into $D$. This is a vertex cover of $G_{\mathrm{bad}}$. For an off-hole centre its cost satisfies

$$r(x) - 1 \le (r(x)-1)(m - r(x)),$$

and for a hole centre it satisfies

$$r(x) - 1 \le r(x)(m - r(x)).$$

Summing and applying (5) gives $|D| \le m\Delta_{\mathcal{H}}(A)$. Any two surviving secants with disjoint endpoint pairs meet at a concurrence centre of index at least two. That centre cannot have index below $m$, since all but one of its secants would then meet $D$; hence its index is $m$. $\square$

**Corollary 3.9** (Matching-design nonexistence gives a two-unit gap). *Let $k \ge 4$, put $m = \lfloor k/2 \rfloor$, and suppose that*

$$v = \frac{3\binom{k}{4}}{\binom{m}{2}}$$

*is an integer. If no simple* $\mathrm{MATCH}(k, m, 1)$ *design exists, then every $k$-arc and every prescribed hole set disjoint from it satisfy*

$$\Delta_{\mathcal{H}}(A) \ge 2.$$

*Proof.* The maximum-index concurrence cliques form a packing of $b$ maximum matchings in $KG(k, 2)$. If $B$ is the number of Kneser edges in the remaining concurrence cliques, the exact concurrence decomposition gives

$$B = (v-b)\binom{m}{2}.$$

Corollary 3.8 gives $B \le \binom{m}{2}\Delta_{\mathcal{H}}(A)$, so $v - b \le \Delta_{\mathcal{H}}(A)$.

It remains to exclude $v - b = 1$. The leave of such a packing has $\binom{m}{2}$ edges. Every positive leave degree is divisible by $m-1$: the ambient Kneser degree and the degree removed by each $K_m$ block are both divisible by $m-1$. Hence the leave has at most $m$ nonisolated vertices, while its minimum positive degree forces at least $m$. It is therefore a $K_m$, supported on $m$ pairwise disjoint edges of $K_k$, and adjoining that maximum matching completes the design. Nonexistence thus forces $v - b \ge 2$, proving the claim. $\square$

*Remark* 3.10 (Slack in the deletion bound). The proof records why the vertex-cover bound can be strict. Include hole points of index 1 with zero deletion cost, let $\mathcal{B}_R$ be the off-hole centres with $2 \le r < m$, and let $\mathcal{B}_H$ be the hole points with $1 \le r < m$. If $D$ is the cover constructed above and

$$\Omega = \sum_{z \in \mathcal{B}_R \cup \mathcal{B}_H} (r(z) - 1) - |D|,$$

then $\Omega \ge 0$ measures overlap among the local deletion sets, and

$$\begin{aligned} m\Delta_{\mathcal{H}}(A) - |D| = \Omega + \sum_{x \in \mathcal{B}_R} (r(x) - 1)(m - r(x) - 1) \\ + \sum_{y \in \mathcal{B}_H} (r(y)(m - r(y)) - (r(y) - 1)). \end{aligned}$$

Thus equality for this cover requires no intermediate-index hole point, requires every bad off-hole centre to have index $m - 1$, and requires the local deletion sets to be disjoint. The identity does not assert that these necessary local conditions are globally realizable.

The line-hole case now makes the connection with complete affine arcs explicit.

**Corollary 3.11** (Complete affine arcs)**.** *Let $L_\infty$ be a line in a projective plane of order $q$, and let $A$ be a $k$-arc disjoint from $L_\infty$. Then $A$ is complete as an arc in the affine plane $\Pi \setminus L_\infty$ if and only if it is complete outside $L_\infty$. In that case*

$$q^2 - k \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{1}{m}\binom{k}{2}. \tag{10}$$

*Equality holds if and only if every affine point outside $A$ has secant index in $\{1, m\}$ and every point of $L_\infty$ has secant index in $\{0, m\}$. If moreover $k \ge 6$ is even and equality holds, then*

$$q \in \left\{ k - 2, \binom{k-1}{2} + 1 \right\}.$$

*Proof.* Maximality in the affine plane is equivalent to every affine point outside $A$ lying on a secant of $A$, which is exactly completeness outside $L_\infty$. Every secant meets $L_\infty$ once, so $I_{L_\infty}(A) = \binom{k}{2}$. Now apply Corollaries 3.3 and 3.2; the latter also gives the equality criterion. For even $k$, substituting the fixed value of $I_{L_\infty}(A)$ into the zero-defect equation and factoring gives

$$(q - (k-2)) \left( q - \left( \binom{k-1}{2} + 1 \right) \right) = 0.$$

□

## 4 Specialization to a conic

Let $\mathcal{C}$ be a nonsingular conic in $\mathrm{PG}(2, q)$, and let $A \subseteq \mathrm{PG}(2, q) \setminus \mathcal{C}$ be a $k$-arc. Since $|\mathcal{C}| = q + 1$, there are $q^2 - k$ points outside $A \cup \mathcal{C}$. Write

$$I_{\mathcal{C}}(A) = \sum_{y \in \mathcal{C}} r(y) = \sum_{\ell \text{ an } A\text{-secant}} |\ell \cap \mathcal{C}|.$$

**Corollary 4.1** (Conic-loss and uncovered-locus inequalities)**.** *For every such arc,*

$$|\mathcal{U}_{\mathcal{C}}(A)| \ge q^2 - k - \binom{k}{2}(q-1) + \frac{6}{m}\binom{k}{4} + \frac{I_{\mathcal{C}}(A)}{m}. \tag{11}$$

*If $A$ is $\mathcal{C}$-complete, then*

$$\boxed{q^2 - k \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{C}}(A)}{m}.} \tag{12}$$

*Proof.* Apply Corollary 3.2 with $\mathcal{H} = \mathcal{C}$, using $|V_{\mathcal{C}}(A)| = q^2 - k$ and $I_{\mathcal{H}}(A) = I_{\mathcal{C}}(A)$. If $A$ is $\mathcal{C}$-complete, then $\mathcal{U}_{\mathcal{C}}(A) = \varnothing$, and the first inequality rearranges to (12). $\square$

The first correction in (12) accounts for unavoidable concurrence among secants; the last accounts for secant incidence spent on the exempt conic. Dropping the last term recovers Corollary 3.4; conic geometry enters the equality theory and finite applications, not this scalar lower bound.

**Corollary 4.2** (Even-size equality spectrum)**.** *Let $k \geq 6$ be even. If a $\mathcal{C}$-complete $k$-arc has zero defect, then*

$$q \in \left\{ k-2,\ \binom{k-1}{2},\ \binom{k-1}{2}+1 \right\}.$$

*In the first two cases every point of $\mathcal{C}$ has index $k/2$; in the third, exactly $k-1$ conic points have that index.*

*Proof.* Put $Q = k-2$ and let $Z = \{x \notin A : r(x) = m\}$. Then $m = Q/2+1$ and, by Theorem 3.6, $|Z| = Q^2 - 1$. Put $s = |Z \cap \mathcal{C}|$. Every positive conic index equals $m$, so $I_{\mathcal{C}}(A) = ms$. Substitution in the zero-defect identity and factorization give

$$s = q + 1 - \frac{(q-Q)(2q - Q(Q+1))}{2}.$$

The elementary arc bound $k \leq q+2$ gives $q \geq Q$. Since $0 \leq s \leq q+1$, either $q = Q$ or $q \geq B := Q(Q+1)/2$. Write $q = B + t$ in the latter case. Then

$$s = q + 1 - t(q - Q).$$

For $t \geq 2$, already at $t = 2$ the subtracted term exceeds $q+1$ by

$$B - 2Q + 1 = \frac{(Q-1)(Q-2)}{2} > 0,$$

and the excess increases with $t$. Hence $t \in \{0, 1\}$. The displayed values of $s$ at $q = Q, B, B+1$ are respectively $q+1, q+1, Q+1$. $\square$

**Corollary 4.3** (Odd-size equality spectrum)**.** *Let $k \geq 7$ be odd. If a $\mathcal{C}$-complete $k$-arc has zero defect, then*

$$q \in \left\{ k-1,\ \binom{k-1}{2} - 1,\ \binom{k-1}{2} \right\}.$$

*If $s$ is the number of conic points of index $(k-1)/2$, then the corresponding values of $s$ are respectively $q, q, k-1$.*

*Proof.* Put $Q = k-1$, $m = Q/2$, and $R = \binom{Q}{2} - 1$. At zero defect every positive conic index equals $m$, and substitution in the defect identity gives

$$s = q - (q-Q)(q-R).$$

The arc bound gives $q \geq Q-1$, but the right-hand side is negative at $q = Q-1$, so $q \geq Q$. If $Q \leq q \leq R$, then $s \leq q+1$ gives $(q-Q)(R-q) \leq 1$. Since $R - Q > 2$, this forces $q = Q$ or $q = R$. If $q = R + t$ with $t \geq 1$, then $s = q - t(q-Q)$. The value $t = 1$ is possible, whereas $t \geq 2$ and $s \geq 0$ would imply $q \leq 2Q$, contrary to $q \geq R + 2 > 2Q$. Substitution gives the stated values of $s$. $\square$

**Corollary 4.4** (Odd equality in characteristic two). *Let $q$ be a power of two, let $k \geq 7$ be odd, and let $A$ be a $k$-arc disjoint from $\mathcal{C}$. Then $A$ is $\mathcal{C}$-complete with zero defect if and only if $k = q+1$, $A$ is an oval, and its nucleus belongs to $\mathcal{C}$.*

*Proof.* Put $Q = k-1$. The value $\binom{Q}{2}$ in Corollary 4.3 is not a power of two: the coprime factors $Q/2$ and $Q-1 > 1$ would both have to be powers of two. Nor can $2^n = \binom{Q}{2} - 1$. Writing $Q = 2a$ and $x = 4a-1$ would give

$$(x-3)(x+3) = 2^{n+3};$$

the two positive powers of two differ by six, hence are 2 and 8, which gives $x = 5$, incompatible with integral $a$. Thus $q = Q$.

A $(q+1)$-arc in even order is an oval. Its nucleus is the unique external point of secant index zero, and every other external point has index $q/2$. The value $s = q$ from Corollary 4.3 therefore says exactly that the prescribed conic contains the nucleus. Conversely, an oval disjoint from $\mathcal{C}$ whose nucleus lies on $\mathcal{C}$ has the required index pattern, so it is $\mathcal{C}$-complete with zero defect. ☐

**Corollary 4.5** (Characteristic-two equality spectrum). *Let $k \geq 6$ be even, let $q$ be a power of two, and suppose that a $\mathcal{C}$-complete $k$-arc has zero defect. Then $k = q+2$.*

*Proof.* Put $Q = k-2$ and $B = Q(Q+1)/2$. By Corollary 4.2, $q \in \{Q, B, B+1\}$. Write $q = 2^n$. The case $q = B$ is impossible: since $Q$ is even, the coprime factors $Q$ and $Q+1$ could have product a power of two only if $Q+1 = 1$.

Suppose $q = B+1$, and write $k = 2m$. Then

$$q = (m-1)(2m-1) + 1.$$

Since $q > 1$ is even, $m$ is even. The last clause of Corollary 4.2 gives $s = k-1$ conic points of index $m$, and hence

$$I_{\mathcal{C}}(A) = ms = \binom{k}{2}.$$

Let $\nu$ be the nucleus of $\mathcal{C}$. If $\nu \in A$, then the $k-1$ lines from $\nu$ to the other arc points are exactly the tangent $A$-secants: any further tangent secant would contain $\nu$ and two other arc points. All nontangent secants meet $\mathcal{C}$ in zero or two points, so $I_{\mathcal{C}}(A) \equiv k-1 \pmod 2$, impossible because $m$ is even. Thus $\nu \notin A$. Now the tangent $A$-secants are exactly the secants through $\nu$, and therefore $I_{\mathcal{C}}(A) \equiv r(\nu) \pmod 2$. Zero defect gives $r(\nu) \in \{1, m\}$, so $r(\nu) = m$. Hence an arc secant $PQ$ is tangent to $\mathcal{C}$, say at $T$.

After a projective change of coordinates, assume that $\mathcal{C}$ is the standard conic. Each point $R \notin \mathcal{C}$ induces an involution $\sigma_R$ on $\mathcal{C}$ by taking the other point on the chord through $R$. Let $S$ be the set of maximum-index conic points. This set is invariant under every $\sigma_R$ with $R \in A$. Indeed, if $Y \in S$, the secants through $Y$ form a perfect matching of $A$, so $R$ has a partner $R' \in A$. The other conic point on $RR'$ is $\sigma_R(Y)$; it lies on an $A$-secant, and zero defect forces its positive conic index to be $m$.

The involutions $\sigma_P$ and $\sigma_Q$ fix $T$. In the standard conic parameter this can be seen directly. Move $T$ to $(1,0,0)$ on $XZ = Y^2$, and write the other conic points as $c(u) = (u^2, u, 1)$. The tangent at $T$ is $Z = 0$, and the chord through $c(u)$ and $c(v)$ meets it at $(u+v, 1, 0)$. Thus a centre $R = (a,1,0)$ induces $\sigma_R : u \mapsto u + a$. The value $a = 0$ is the nucleus and gives the identity; since $P, Q \in A$, the preceding exclusion of the nucleus and $P \neq Q$ give distinct nonzero parameters $a_P, a_Q$. Hence $\sigma_P, \sigma_Q$, and their product are the three nonidentity translations with parameters $a_P, a_Q, a_P + a_Q$. They commute and each has $T$ as its unique fixed point. All three preserve $S$, and

$$|S| = k - 1 \equiv 3 \pmod 4.$$

Restricted to $S$, each nonidentity involution has one fixed point and therefore an odd number of transpositions. Thus all three permutations have sign $-1$, contradicting $\operatorname{sgn}(\sigma_P\sigma_Q) = \operatorname{sgn}(\sigma_P)\operatorname{sgn}(\sigma_Q) = 1$. The case $q = B + 1$ is therefore impossible. The remaining case $q = Q$ is $k = q + 2$. □

In the first branch, $A$ is a hyperoval. Every point outside $A$ has secant index $(q+2)/2$, so every nonsingular conic disjoint from $A$ produces a $\mathcal{C}$-complete zero-defect pair. Thus the oval and hyperoval branches in characteristic two are completely geometric.

**Corollary 4.6** (Small even equality cases). *No zero-defect $\mathcal{C}$-complete eight- or twelve-arc exists in a Desarguesian plane. A zero-defect ten-arc can occur only in* $\mathrm{PG}(2, 8)$*; its underlying arc has the regular-hyperoval concurrence design, and such equality pairs exist. This statement does not classify either their projective realizations or the prescribed conics disjoint from them.*

*Moreover, if $Q \geq 4$ is an even prime power and $d \geq 2$, no $\mathcal{C}$-complete $(Q+2)$-arc in* $\mathrm{PG}(2, Q^d) \setminus \mathcal{C}$ *has zero defect.*

*Proof.* For $k = 8, 10, 12$, the three candidate orders are respectively

$$\{6, 21, 22\}, \qquad \{8, 36, 37\}, \qquad \{10, 55, 56\}.$$

Since a Desarguesian plane has prime-power order, this excludes $k = 8, 12$ and leaves only $q = 8, 37$ when $k = 10$.

For the final assertion,

$$Q^d \geq Q^2 > \frac{Q(Q+1)}{2} + 1$$

when $Q \geq 4$ and $d \geq 2$, so none of the three orders allowed by Corollary 4.2 equals $Q^d$.

For the ten-point refinement, Theorem 3.6 produces a rank-three MATCH(10, 5, 1) design. Apply Theorem A.3.

For existence over $\mathbb{F}_8 = \mathbb{F}_2(\alpha)$, where $\alpha^3 + \alpha + 1 = 0$, distinguish the constituent conic

$$\mathcal{D}: \; XZ = Y^2$$

from the prescribed conic, and denote its nucleus by $N_{\mathcal{D}} = (0, 1, 0)$. Let

$$A = \mathcal{D} \cup \{N_{\mathcal{D}}\}, \qquad \mathcal{C}: \; X^2 + Y^2 + Z^2 + (1+\alpha)YZ = 0.$$

The gradient of the latter quadratic is $(0, (1+\alpha)Z, (1+\alpha)Y)$, so $\mathcal{C}$ is nonsingular. Its values on the eight affine points $(t^2, t, 1)$ of $A$, in the polynomial-basis order $t = 0, 1, \alpha, 1 + \alpha, \ldots, 1 + \alpha + \alpha^2$, are

$$1, \; \alpha, \; 1+\alpha^2, \; \alpha+\alpha^2, \; \alpha, \; 1, \; \alpha+\alpha^2, \; 1+\alpha^2,$$

and its value at each of the two infinite points of $A$ is 1. Hence $A \cap \mathcal{C} = \varnothing$. Every point outside a hyperoval in $\mathrm{PG}(2, 8)$ lies on five of its secants: the ten hyperoval points pair on the secants through that point. Thus every point of $\mathcal{C}$ has index five, so $A$ is $\mathcal{C}$-complete and has zero defect. □

**Corollary 4.7** (Explicit integer-valued lower bound)**.** *Define*

$$L_1(q) = \min\left\{k \geq 3 : q^2 - k \leq \binom{k}{2}(q-1)\right\},$$
$$L_2(q) = \min\left\{k \geq 3 : q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{\lfloor k/2 \rfloor}\binom{k}{4}\right\}.$$

*Then*

$$\rho_{\mathcal{C}}(q) \geq L_2(q) \geq L_1(q). \tag{13}$$

*Proof.* For $q \geq 3$ a set of at most two points has at most one secant, and that secant together with the set and the conic contains at most $2q + 2 < q^2 + q + 1$ points. Thus a $\mathcal{C}$-complete arc has at least three points, so the moment bound applies to the unrestricted minimum defining $\rho_{\mathcal{C}}(q)$. The definition of $L_2(q)$ involves only integer arithmetic after multiplication by $\lfloor k/2 \rfloor$. The correction subtracted in the definition of $L_2(q)$ is nonnegative, so every integer feasible for $L_2(q)$ is feasible for $L_1(q)$; hence $L_2(q) \geq L_1(q)$. □

For calculation, the corrected capacity has the parity-dependent forms

$$\binom{k}{2}(q-1) - \frac{6}{\lfloor k/2 \rfloor}\binom{k}{4} = \begin{cases} \dfrac{k-1}{2}\big(k(q-1) - (k-2)(k-3)\big), & k \text{ even}, \\ \dfrac{k}{2}\big((k-1)(q-1) - (k-2)(k-3)\big), & k \text{ odd}. \end{cases} \tag{14}$$

For comparison, some values are

| $q$ | 5 | 8 | 9 | 11 | 13 | 16 | 17 | 19 |
|---|---|---|---|---|---|---|---|---|
| $L_1(q)$ | 4 | 5 | 5 | 6 | 6 | 7 | 7 | 7 |
| $L_2(q)$ | 4 | 6 | 6 | 6 | 7 | 8 | 8 | 8 |
| $\rho_{\mathcal{C}}(q)$ | 4 | 6 | 6 | 6 | 8 | 9 | 9 | 10 |

The last row combines the analytic lower bounds at $q = 5, 8, 9, 11$, the kernel-checked classification at $q = 16$, and the exhaustive classifications at $q = 13, 17, 19$ described in Section D. Thus the second-moment correction alone proves, for example, that a $\mathcal{C}$-complete arc in $\mathrm{PG}(2, 16)$ has at least eight points, while the finite classifications determine the displayed exact values.

The routine algebra behind the additive bound is isolated in the next lemma on the full interval needed to avoid any parity split.

**Lemma 4.8** (Polynomial estimate)**.** *For $s \geq 2$ and $0 \leq a \leq 2$, put*

$$P_s(a) = \frac{2a-3}{4}s^3 + \frac{a^2 - 7a + 10}{4}s^2 + \frac{-3a^2 + 10a - 8}{2}s + \frac{-a^3 + 5a^2 - 8a + 6}{2}.$$

*If $P_s(a) \geq 0$, then $a \geq 3/2 - 8/s$.*

*Proof.* On $0 \leq a \leq 2$, the coefficients of $s^2$, $s$, and 1 in $P_s(a)$ are at most $5/2$, $1/6$, and 3, respectively; for the middle coefficient, $1/6 - (-3a^2 + 10a - 8)/2 = (3a - 5)^2/6$. Since $s \geq 2$, their total contribution is at most

$$\frac{5}{2}s^2 + \frac{1}{6}s + 3 \leq 4s^2.$$

Thus $P_s(a) \geq 0$ implies $0 \leq (2a - 3)s^3/4 + 4s^2$, and division by $s^2$ gives the claim. □

**Theorem 4.9** (Explicit additive lower bound)**.** *For every prime power $q$,*

$$\boxed{\rho_{\mathcal{C}}(q) \geq \sqrt{2q} + \frac{3}{2} - \frac{8}{\sqrt{2q}}.} \tag{15}$$

*Consequently, as $q$ tends to infinity through prime powers,*

$$\liminf_{q\to\infty}(\rho_{\mathcal{C}}(q) - \sqrt{2q}) \geq \frac{3}{2}. \tag{16}$$

*Proof.* For $q = 2$ the right side of (15) is negative, so the assertion is immediate. Assume $q \geq 3$. Let $k$ be the size of a $\mathcal{C}$-complete arc and put $s = \sqrt{2q}$. The elementary first-moment inequality $q^2 - k \leq \binom{k}{2}(q-1)$ implies $k \geq s$ for $q \geq 2$: otherwise $k < s \leq q$, while $k^2 < 2q$ gives $\binom{k}{2} < q$, so its right side is smaller than $q(q-1)$ whereas $q^2 - k \geq q(q-1)$. If $k \geq s+2$, then (15) is immediate. Otherwise write $k = s + a$, so $0 \leq a < 2$ and $s \geq 2$.

Since $m = \lfloor k/2 \rfloor \leq k/2$,

$$\frac{6}{m}\binom{k}{4} \geq \frac{12}{k}\binom{k}{4} = \frac{(k-1)(k-2)(k-3)}{2}.$$

Dropping the nonnegative conic term in (12) therefore gives the necessary inequality

$$q^2 - k \leq \frac{k-1}{2}\big(k(q-1) - (k-2)(k-3)\big).$$

After substituting $q = s^2/2$ and $k = s + a$, the right side minus the left side is

$$\begin{aligned}&\frac{s+a-1}{2}\left((s+a)\left(\frac{s^2}{2}-1\right) - (s+a-2)(s+a-3)\right) + (s+a) - \frac{s^4}{4}\\ &\qquad= \frac{2a-3}{4}s^3 + \frac{a^2-7a+10}{4}s^2 + \frac{-3a^2+10a-8}{2}s + \frac{-a^3+5a^2-8a+6}{2} = P_s(a).\end{aligned}$$

It is nonnegative, so Lemma 4.8 gives $a \geq 3/2 - 8/s$, which is (15). Letting $q$ tend to infinity through prime powers gives (16). ☐

*Remark* 4.10 (The scale of the conic term)*.* Every secant meets $\mathcal{C}$ in at most two points, so

$$0 \leq I_{\mathcal{C}}(A) \leq 2\binom{k}{2}.$$

At the relevant scale $k \asymp \sqrt{q}$, the term $I_{\mathcal{C}}(A)/m$ is therefore $O(\sqrt{q})$, while the universal overlap correction $6\binom{k}{4}/m$ is $\Theta(q^{3/2})$. Consequently, improving the lower bound on $I_{\mathcal{C}}(A)$ within (12) can be useful for finely balanced finite cases, but cannot improve the additive constant $3/2$ in Theorem 4.9. Any stronger asymptotic result needs an additional mechanism, not merely a spectral estimate for $I_{\mathcal{C}}(A)$.

# 5 Conclusion and further questions

The prescribed-hole identity retains the local equality and stability data discarded by the usual scalar secant count. At zero defect those local conditions assemble into a matching design. For any $q+1$ prescribed holes they sharpen the scalar lower bound; for a conic, the same mechanism also restricts equality and isolates the finite obstruction at $q = 16$. The appendices record secondary consequences and the exact formal and computational trust boundaries.

**Problem 5.1.** Construct $\mathcal{C}$-complete arcs of size $O(\sqrt{q})$, or prove that this is impossible for an infinite family. The upper-bound transfer in Appendix B currently supplies only $O(\sqrt{q}(\log q)^c)$. On the lower-bound side, Remark 4.10 shows that improving the additive constant $3/2$ requires a mechanism coupling rank-three realization to the matching structure, rather than a sharper estimate of the conic-incidence term.

**Problem 5.2.** Develop the higher-dimensional analogue for caps complete outside a prescribed quadric in $\mathrm{PG}(n, q)$. The evaluation lemma already applies to arbitrary feature maps and Veronese degrees; the missing ingredient is an appropriate exact secant-moment remainder for caps.

# A Equality classifications

The first two nontrivial orders illustrate the distinction between an abstract matching design and its rank-three projective realization. This secondary classification is not needed for the defect identity, the conic lower bound, or the $q = 16$ exclusion. Its role is to show that abstract matching-design existence is weaker than projective realizability, which can impose field and characteristic restrictions.

**Definition A.1.** A *rank-three projective realization* of a $\mathrm{MATCH}(k, m, 1)$ design $\mathcal{D}$ over a field $K$ is an injective assignment $i \mapsto p_i \in \mathrm{PG}(2, K)$ whose image is a $k$-arc and such that, for every matching $M \in \mathcal{D}$, the $m$ lines $p_i p_j$, $\{i, j\} \in M$, are concurrent. Two matching designs are *isomorphic* when a permutation of their $k$ vertices carries the matching collection of one onto that of the other.

**Corollary A.2** (Six and seven points)**.** *(a) In a Desarguesian plane, if a six-arc has every intersection of two disjoint secants on a third secant, then the field has characteristic two, contains $\mathbb{F}_4$, and the arc is projectively equivalent to*

$$(1,0,0),\ (0,1,0),\ (0,0,1),\ (1,1,1),\ (1,\omega,\omega^2),\ (1,\omega^2,\omega), \qquad \omega^2+\omega+1=0.$$

*Conversely, this configuration has the stated concurrence property. In particular, every zero-defect six-arc in a Desarguesian plane has this form.*

*(b) In any projective plane, every seven-arc has $\Delta_{\mathcal{H}}(A) \geq 2$ with respect to every prescribed hole set.*

*Proof.* For six points, the three intersections of opposite sides of any complete quadrangle must all lie on the secant through the remaining pair. Normalize four vertices to

$$(1,0,0), \quad (0,1,0), \quad (0,0,1), \quad (1,1,1).$$

Their diagonal points are

$$(1,1,0), \qquad (1,0,1), \qquad (0,1,1).$$

Their determinant is $-2$, so they are collinear exactly in characteristic two, on $x+y+z=0$; that line contains the remaining pair. Write a fifth point as $(1, t, 1+t)$. Applying the diagonal-line condition to the quadrangle formed by the first three frame points and this fifth point gives the diagonal line

$$t(1+t)x + (1+t)y + tz = 0.$$

It must contain the remaining frame point $(1,1,1)$, which gives $t^2+t+1=0$; its intersection with $x+y+z=0$ is the sixth point $(1,t^2,t)$. Direct substitution verifies all required concurrences.

For seven points, Theorem 3.6 would produce a MATCH$(7,3,1)$ design. Alspach and Heinrich [3, Theorem 3.1] proved that no such design exists. Here $v=35$, so Corollary 3.9 gives the stated quantitative bound. □

**Theorem A.3** (Rank-three ten-point classification)**.** *Let $K$ be a field. Among the simple* MATCH$(10,5,1)$ *designs, only the regular-hyperoval design can have a rank-three projective realization. It has one over $K$ if and only if*

$$\operatorname{char} K = 2, \qquad \textit{and} \qquad \mathbb{F}_8 \subseteq K.$$

*The other abstract design has no rank-three realization over any field.*

*Proof.* Alspach and Heinrich allow repeated matchings in their definition, but when $\lambda=1$ a repeated five-matching would repeat each of its pairs of edges and is therefore impossible. Thus their statement that Mathon found precisely two nonisomorphic MATCH$(10,5,1)$ designs applies to Definition A.1 [3, p. 40, opening classification paragraph]. Equivalently, these are the two partial geometries denoted pg$(5,7,3)$ in Reichard and Woldar's (line size, point degree, $\alpha$) convention, or pg$(4,6,3)$ in the common pg$(s,t,\alpha)$ convention. Reichard and Woldar identify Mathon's result as a full classification and construct models of both classes in Proposition 5.1.5 and Corollary 5.1.6 [16, Proposition 5.1.5 and Corollary 5.1.6]. This published two-class completeness is an external input. The paper-local enumeration in Appendix F reconstructs and checks representatives of the two classes, but is not used to prove that the list is complete.

Here is the realization test. Relabel and normalize four vertices to

$$p_0=(1,0,0), \quad p_1=(0,1,0), \quad p_2=(0,0,1), \quad p_3=(1,1,1).$$

No remaining point lies on $p_0p_1$, so write $p_i=(x_i,y_i,1)$ for $4\le i\le 9$. For each matching block, the first two secants determine its proposed concurrence point; requiring each of the other three secants to pass through that point gives three determinant equations. The 63 blocks therefore give 189 integral equations, and together with the arc inequations they are necessary and sufficient for a realization.

Only a necessary part of the arc open set is needed for the exclusions. Indeed,

$$\det(p_1,p_2,p_9)=x_9, \qquad \det(p_1,p_3,p_9)=x_9-1,$$

so every arc realization has $x_9(x_9-1)\neq 0$. The certificate saturates the concurrency ideal by this product. This enlarges, rather than shrinks, the locus that must be excluded: a unit saturated ideal therefore rules out every arc realization without claiming that this one product encodes all arc inequations.

The remaining elimination is computer-assisted. The exact certificate summarized in Appendix F.2 gives the following mathematical output. After inverting 2, rational module identities make both saturated ideals the unit ideal, and their only denominator prime is 2; hence both designs are excluded in every odd characteristic. In characteristic two the second design again has unit saturated ideal. For the regular-hyperoval design, the surviving triangular basis forces a parameter $t$ with $t^3+t+1=0$, while the arc inequations force $t\notin\mathbb{F}_2$. Thus $\mathbb{F}_8\subseteq K$. The regular hyperoval over $\mathbb{F}_8$ and scalar extension prove the converse. □

## B An upper-bound transfer from complete arcs

Let $t_2(2,q)$ denote the smallest size of an ordinary complete arc in $\mathrm{PG}(2,q)$.

**Theorem B.1** (Projective averaging). *Let $H \subseteq \mathrm{PG}(2,q)$ and let $B$ be an ordinary complete $b$-arc. If*

$$b|H| < q^2 + q + 1, \tag{17}$$

*then some projective image of $B$ is disjoint from $H$ and complete outside $H$.*

*Proof.* Choose a uniformly random projectivity $g \in \mathrm{PGL}(3,q)$. Point transitivity gives

$$\mathbb{E}|gB \cap H| = \frac{b|H|}{q^2+q+1} < 1.$$

Since the intersection size is a nonnegative integer, it is zero for some $g$. Projectivities preserve ordinary completeness, so $gB$ is complete outside $H$. □

**Corollary B.2** (Conic transfer). *If $t_2(2,q) \le q$, then*

$$\rho_{\mathcal{C}}(q) \le t_2(2,q). \tag{18}$$

*More generally, every complete $b$-arc with $b \le q$ has a projective image disjoint from the prescribed conic.*

*Proof.* Take $H = \mathcal{C}$, so $|H| = q+1$. For integral $b \le q$ one has $b(q+1) < q^2+q+1$, and Theorem B.1 applies. □

The hypothesis $t_2(2,q) \le q$ is part of the corollary, not a uniform assertion about every order. The application below uses only the range of sufficiently large $q$, where the Kim–Vu bound supplies it.

Kim and Vu proved that, for all sufficiently large $q$, every projective plane of order $q$ has a complete arc of size at most $\sqrt{q}(\log q)^c$ for an absolute constant $c$ [11, Theorem 1.1]. This is less than $q$ for large $q$, so Corollary B.2 gives the following immediate consequence.

**Corollary B.3.** *For some absolute constant $c$ and all sufficiently large prime powers $q$,*

$$\rho_{\mathcal{C}}(q) \le \sqrt{q}(\log q)^c.$$

The transfer does not exploit the conic and is unlikely to be sharp. It nevertheless supplies a uniform upper bound without having to preserve the arc condition during a separate random covering construction.

## C Even characteristic and the nucleus

Assume $q$ is even and let $\nu$ be the nucleus of $\mathcal{C}$. The nucleus is outside $\mathcal{C}$, all tangents to $\mathcal{C}$ pass through $\nu$, and every line through $\nu$ is one of these $q+1$ tangents.

**Proposition C.1** (Case $\nu \in A$). *Let $A$ be a $k$-arc disjoint from $\mathcal{C}$ with $\nu \in A$. Exactly $k-1$ $A$-secants are tangent to $\mathcal{C}$. In particular,*

$$I_{\mathcal{C}}(A) \ge k-1, \qquad I_{\mathcal{C}}(A) \equiv k-1 \pmod 2.$$

*If $A$ is $\mathcal{C}$-complete, then*

$$q^2 - k \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{k-1}{m}. \tag{19}$$

*Proof.* For every $a \in A \setminus \{\nu\}$, the line $\nu a$ is tangent to $\mathcal{C}$, giving $k-1$ tangent secants. No line joining two points of $A \setminus \{\nu\}$ can be tangent: every tangent contains $\nu$, which would place three points of $A$ on one line. Hence these are exactly the tangent $A$-secants.

Every tangent contributes one to $I_{\mathcal{C}}(A)$, while every other line contributes zero or two. This proves the lower bound and parity statement. Inequality (19) follows from Corollary 4.1. □

**Proposition C.2** (Case $\nu \notin A$)**.** *Let $A$ be a $\mathcal{C}$-complete arc with $\nu \notin A$. Then*

$$1 \leq r(\nu) \leq m,$$

*the tangent $A$-secants are exactly the secants through $\nu$, and*

$$I_{\mathcal{C}}(A) \equiv r(\nu) \pmod 2, \qquad I_{\mathcal{C}}(A) \geq r(\nu) \geq 1.$$

*Consequently,*

$$q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{1}{m}. \tag{20}$$

*Proof.* The nucleus is a required point outside $A \cup \mathcal{C}$, so relative completeness gives $r(\nu) \geq 1$; Lemma 2.1 gives the upper bound. A line is tangent to $\mathcal{C}$ exactly when it passes through $\nu$. Thus the number of tangent $A$-secants is $r(\nu)$. Reducing the sum of the intersection sizes $|\ell \cap \mathcal{C}|$ modulo two gives $I_{\mathcal{C}}(A) \equiv r(\nu) \pmod 2$. The remaining claims follow from Corollary 4.1. □

**Corollary C.3** (Universal even-characteristic loss)**.** *Every $\mathcal{C}$-complete arc in even characteristic satisfies*

$$I_{\mathcal{C}}(A) \geq 1.$$

*Consequently, inequality* (12) *can always be strengthened by replacing its final term by at least $-1/m$.*

*Proof.* If the nucleus belongs to $A$, Proposition C.1 gives $I_{\mathcal{C}}(A) \geq k-1 \geq 1$. Otherwise Proposition C.2 gives $I_{\mathcal{C}}(A) \geq r(\nu) \geq 1$. □

These two cases are useful structural reductions, but Remark 4.10 limits what the incidence term alone can accomplish. For example, at $(q,k) = (16,8)$ inequality (12) would be contradicted only by $I_{\mathcal{C}}(A) > 268$, whereas the universal upper bound is $I_{\mathcal{C}}(A) \leq 56$. Thus the exact determination of $\rho_{\mathcal{C}}(16)$ below requires information beyond a lower bound on $I_{\mathcal{C}}(A)$ inserted into the present inequality.

# D Verified finite-field examples

The verified values at $q = 5, 8, 9, 11$ use the lower bound $L_2(q)$ and explicit coordinates. At $q = 5$ the witness is the projective four-frame. Exhaustive projective classifications reject size 7 at $q = 13$, size 8 at $q = 17$, and sizes $8, 9$ at $q = 19$; checked witnesses attain sizes $8, 9, 10$. At $q = 16$, the defect bound and a checked witness give $8 \leq \rho_{\mathcal{C}}(16) \leq 9$; a checked exhaustive augmentation covering list excludes eight. The proof has three steps: the defect bound supplies the lower bound eight; a hypothetical $\mathcal{C}$-complete eight-arc would have its ordinary-uncovered locus on a quadratic avoiding the arc; and the certificate rules out every such quadratic, while the displayed nine-point witness supplies the upper bound.

## D.1 Evaluation obstruction

The algebraic rejection used in that classification is an instance of the following elementary linear-algebra observation. Linear systems of quadrics containing arcs are an established tool; see Glynn [9] and Ball [5]. Fix nonzero vector representatives of the projective points under consideration. If $V$ is a finite-dimensional linear system of homogeneous forms, write $\mathrm{ev}_x \in V^*$ for evaluation at the representative of $x$. The zero or nonzero status of this evaluation is independent of the chosen representative.

Indeed, if an arc is complete outside a conic, then its ordinary-uncovered locus lies in the zero set of the conic equation while every selected point avoids that zero set. The next lemma isolates exactly the linear-algebraic obstruction used to rule this out at $q = 16$.

**Lemma D.1** (Uncovered evaluation obstruction)**.** *Let $A$ and $U$ be finite sets of projective points. There is no nonzero $f \in V$ such that $U \subseteq Z(f)$ and $A \cap Z(f) = \varnothing$ if either of the following holds:*

1. *the evaluation map $f \mapsto (\mathrm{ev}_u(f))_{u \in U}$ is injective;*
2. *for some $a \in A$, the functional $\mathrm{ev}_a$ belongs to the span of $\{\mathrm{ev}_u : u \in U\}$.*

*Proof.* If $f$ vanishes on $U$, injectivity in the first case gives $f = 0$. In the second case, applying a spanning relation for $\mathrm{ev}_a$ to $f$ gives $f(a) = 0$, contrary to $A \cap Z(f) = \varnothing$. ☐

**Lemma D.2** (Line–triangle obstruction)**.** *Suppose that $U \subseteq \mathrm{PG}(2, K)$ contains three distinct points on a line $L$, together with three noncollinear points outside $L$. Then no nonzero homogeneous quadratic vanishes on $U$.*

*Proof.* The restriction of such a quadratic to $L \cong \mathrm{PG}(1, K)$ has degree two and three distinct zeros, so it vanishes identically. Hence the equation of $L$ divides the quadratic. The residual linear factor must vanish at the three points outside $L$, contradicting their noncollinearity unless that factor, and hence the quadratic, is zero. ☐

For the full system of degree-$d$ forms, the evaluation functionals are the degree-$d$ Veronese images of the points, up to nonzero scalar. Thus the first alternative says that the uncovered Veronese images span the dual coefficient space; the second supplies a selected Veronese image already in their span. In particular, for the full degree-$d$ system it is enough to exhibit $\binom{d+2}{2}$ uncovered points whose Veronese evaluation matrix is invertible. This formulation is not specific to conics.

## D.2 Small values and the classification over $\mathbb{F}_{16}$

**Proposition D.3.** *For the standard conic*

$$\mathcal{C} : \; XZ = Y^2$$

*in the indicated projective planes,*

$$\rho_{\mathcal{C}}(5) = 4, \quad \rho_{\mathcal{C}}(8) = 6, \quad \rho_{\mathcal{C}}(9) = 6, \quad \rho_{\mathcal{C}}(11) = 6,$$
$$\rho_{\mathcal{C}}(13) = 8, \quad \rho_{\mathcal{C}}(17) = 9, \quad \rho_{\mathcal{C}}(19) = 10.$$

*Proof.* Equation (13) gives

$$L_2(5) = 4, \qquad L_2(8) = L_2(9) = L_2(11) = 6.$$

For $q = 5$, the standard four-frame has ordinary-uncovered locus

$$V(X^2 + Y^2 + Z^2 + XY + XZ + YZ),$$

a nonsingular six-point conic. An invertible coordinate change carries this conic to $XZ = Y^2$ and the frame to the four coordinates in Appendix G; the relative certificate is checked in Lean. The appendix also lists $\mathcal{C}$-complete arcs of size six for $q = 8, 9, 11$. It also lists the attaining arcs at $q = 13, 17, 19$; their relative certificates are kernel-checked. For the reverse inequalities, the classifier enumerates all 80 projective seven-arc classes at $q = 13$, all 5441 projective eight-arc classes at $q = 17$, all 20,361 projective eight-arc classes at $q = 19$, and all 1,053,996 legal nine-arc extensions of those classes. In each case the ordinary-uncovered quadratic evaluation matrix has full rank six, excluding containment in any conic avoiding the arc. Appendix F records the exhaustive-search contract and the external-execution boundary. □

The ordinary classification of eight-arcs over $\mathbb{F}_{16}$ records arc counts, stabilizers, secant-index distributions, and conic-contained classes. The theorem below adds the datum needed here: the ordinary uncovered locus either imposes six independent quadratic conditions or forces every containing quadratic to meet the arc.

The proof uses an exhaustive covering list, not pairwise inequivalent representatives. The following contract isolates the mathematical completeness step; Appendix F records the normalized point encoding, generator, certificate checker, and trust boundary. Let $F$ denote the standard projective four-frame.

**Lemma D.4** (Augmentation certificate contract)**.** *For $4 \le i \le 8$, let $\mathcal{L}_i$ be a finite list of $i$-arcs with $\mathcal{L}_4 = \{F\}$. Suppose that for every $4 \le i < 8$, every $P \in \mathcal{L}_i$, and every $x \notin P$ such that $P \cup \{x\}$ is an arc, the certificate supplies a child $P' \in \mathcal{L}_{i+1}$ and an invertible linear map whose induced projectivity carries $P \cup \{x\}$ onto $P'$. Then every eight-arc in* $\mathrm{PG}(2, 16)$ *is projectively equivalent to a member of $\mathcal{L}_8$.*

*Proof.* Choose four points of the eight-arc. Every four-arc is a projective frame, so a projectivity carries those points to $F$. Insert the remaining four points in any order. At each step the stated certificate transports the legal insertion to the next list. Induction through the four layers gives a member of $\mathcal{L}_8$ projectively equivalent to the original arc. □

**Theorem D.5** (Quadratic avoidance over $\mathbb{F}_{16}$)**.** *Let $A$ be any eight-arc in* $\mathrm{PG}(2, 16)$*, and let $U(A)$ be the points outside $A$ that lie on no secant of $A$. There is no nonzero homogeneous quadratic form $Q$ such that*

$$U(A) \subseteq Z(Q) \qquad \text{and} \qquad A \cap Z(Q) = \varnothing.$$

*The assertion includes singular quadratic forms.*

*Proof. Covering list.* The checked list lengths are $|\mathcal{L}_5| = 4$, $|\mathcal{L}_6| = 61$, $|\mathcal{L}_7| = 454$, and $|\mathcal{L}_8| = 2633$. Every legal insertion among all 273 projective points has a checked child and projective transporter, so Lemma D.4 proves exhaustiveness. The final length agrees with the independent class count of Al-Seraji–Al-Ogali [2, Theorem 3.8.1], but that agreement is not used.

*Leaf obstruction.* For 2630 leaves, the uncovered locus contains three collinear points and three noncollinear points outside their line. Lemma D.2 replaces the separate quadratic matrix inversions on all these leaves.

Only three leaves lack this pattern. Using the polynomial-basis integer encoding of Appendix G, their one-dimensional quadratic kernels are generated, up to scalar, by

| leaf | $Q$ | $A \cap Z(Q)$ |
|---|---|---|
| 89 | $X^2 + Y^2 + Z^2 + XY + XZ$ | $\{(1, 8, 14), (1, 11, 12)\}$ |
| 90 | $5XY + 4XZ + YZ$ | 7 of the 8 selected points |
| 2631 | $2X^2 + Y^2 + Z^2 + 5XY + 5XZ + YZ$ | $\{(1, 6, 12), (1, 11, 7)\}$. |

The first and third forms split respectively as

$$(X + 6Y + 6Z)(X + 7Y + 7Z) \quad \text{and} \quad 2(X + 4Y + 15Z)(X + 15Y + 4Z);$$

the middle form is nonsingular. Direct substitution gives the displayed arc intersections. Thus every quadratic containing the uncovered locus meets the arc on each exceptional leaf, which is the second alternative of Lemma D.1. Projective transport preserves the assertion, so the checked leaf result holds for every eight-arc. □

**Corollary D.6** (Exact relative value over $\mathbb{F}_{16}$)**.** *For every nonsingular conic $\mathcal{C} \subseteq \mathrm{PG}(2, 16)$,*

$$\rho_{\mathcal{C}}(16) = 9.$$

*Proof.* Equation (13) gives $L_2(16) = 8$. If a $\mathcal{C}$-complete eight-arc existed, its ordinary-uncovered locus would lie in the zero set of the nonzero quadratic defining $\mathcal{C}$, while the arc is disjoint from that zero set. This contradicts Theorem D.5. The nine-point witness in Appendix G gives the matching upper bound. □

# E An inverse coda: uncovered-locus reconstruction

This coda does not apply to the $\mathcal{C}$-complete arcs studied above: they satisfy $U(A) \subseteq \mathcal{C}$, and the first-moment covering inequality gives $\binom{k}{2} \geq q+1$, opposite to the strict hypothesis below.

For fixed arc length and sufficiently large $q$, the ordinary-uncovered locus remembers its parent arc. The inverse map has two elementary stages. Put $N = \binom{k}{2}$ and $S = \mathrm{PG}(2, q) \setminus U(A)$. First recover the secant arrangement from the gap between $q + 1$, the number of points on a genuine secant, and $N$, the largest number of points in which a nonsecant line can meet the union. Then recover the vertices from the gap between their secant multiplicity $k - 1$ and the largest possible nonvertex multiplicity $\lfloor k/2 \rfloor$:

$$U(A) \longmapsto S \longmapsto \{\ell : |\ell \cap S| > N\} \longmapsto \{x : \deg(x) = k - 1\} = A.$$

The strict first gap is the only large-field hypothesis.

**Proposition E.1** (Uncovered-locus reconstruction)**.** *Let $A$ and $B$ be $k$-arcs in $\mathrm{PG}(2, q)$, where $k \geq 3$ and*

$$q + 1 > \binom{k}{2}.$$

*If $U(A) = U(B)$ as literal subsets of the plane, then $A = B$ as unlabelled point sets.*

*Proof.* The complement of $U(A)$ is the union of the $\binom{k}{2}$ distinct secants of $A$, and likewise for $B$. Let $\ell$ be a secant of $A$. If no secant of $B$ were equal to $\ell$, then the secants of $B$ would cover at most $\binom{k}{2}$ points of $\ell$, one per line. This is impossible because $|\ell| = q + 1 > \binom{k}{2}$ and the two secant unions are equal. Thus every secant of $A$ is a secant of $B$; since the two sets have the same finite size, their secant-line sets coincide.

It remains to recover the vertices from this line arrangement. Each point of $A$ lies on its $k-1$ incident secants. A point outside $A$ lies on at most $\lfloor k/2 \rfloor$ secants: distinct secants through it use disjoint pairs of the $k$ vertices and hence form a matching. Since $k-1 > \lfloor k/2 \rfloor$ for $k \geq 3$, the set $A$ is exactly the set of points incident with $k-1$ recovered secants, and the same description recovers $B$. □

**Corollary E.2** (Canonical inverse and automorphisms)**.** *Under the hypotheses of Proposition E.1, put $S = \mathrm{PG}(2,q) \setminus U(A)$ and $N = \binom{k}{2}$. The secants of $A$ are exactly the lines $\ell$ with $|\ell \cap S| > N$, and $A$ is exactly the set of points incident with $k-1$ such lines. Moreover,*

$$\mathrm{Stab}_{\mathrm{P\Gamma L}_3(q)}(U(A)) = \mathrm{Stab}_{\mathrm{P\Gamma L}_3(q)}(A).$$

*Proof.* Every secant lies in $S$ and has $q+1 > N$ points. A nonsecant line meets each of the $N$ secants in at most one point, so it contains at most $N$ points of their union $S$. The vertex-recovery rule is the one proved above. Finally, every semilinear projectivity stabilizing $A$ stabilizes $U(A)$. Conversely, if $gU(A) = U(A)$, then $U(gA) = gU(A) = U(A)$, so Proposition E.1 gives $gA = A$. □

*Remark* E.3 (Sharpness of the strict threshold). The strict inequality cannot in general be weakened to $q+1 \geq \binom{k}{2}$. For the projective four-frame $A$ over $\mathbb{F}_5$, Proposition D.3 gives $U(A) = \mathcal{C}(\mathbb{F}_5)$, while $q+1 = \binom{4}{2} = 6$. The stabilizer of $\mathcal{C}$ in $\mathrm{PGL}_3(5)$ has order $|\mathrm{PGL}_2(5)| = 120$, whereas the setwise stabilizer of a projective frame has order $4! = 24$. The conic stabilizer therefore moves $A$ through at least five distinct four-frames, all with the same literal uncovered locus $\mathcal{C}(\mathbb{F}_5)$.

# F Computational and formal verification

## F.1 Formal gate and evidence roles

The main body develops the mathematical mechanism; the appendices separate secondary deductions from finite exclusion. The corresponding evidence and trust boundaries are summarized below. In the public `finitegeom` repository, the base human boundary is recorded in `trust/ARCS_COMPLETE_OUTSIDE_CONIC.md`. The matching-packing and small odd-order supplement is recorded separately in `trust/ARCS_COMPLETE_OUTSIDE_CONIC_ADDITIONS.md`. Under `RelativeConicArcs/Gates/`, the corresponding import-only modules are

`ArcsCompleteOutsideConicHuman.lean`,
`ArcsCompleteOutsideConicAdditions.lean`.

The archived `finitegeom` release 0.2.0 has DOI `10.5281/zenodo.21664257`; the later supplement is fixed at public commit `0b3f37d`. The generated order-16 proof is distributed separately in `finitegeom-q16-certificates`, fixed at public commit `ecee482d` and pinned there to `finitegeom` commit `ef6317c5`. This appendix states the mathematical certificate contracts and the trust boundary; raw generated certificates, generators, and replay scripts remain in the companion.

The human gate and supplement also check the discrete gap, complete-affine equality orders, three-value equality spectra, and the standard-conic characteristic-two conclusions. The identification of an arbitrary equality oval or hyperoval with its prescribed-conic configuration remains a paper proof.

The complete `#print axioms` output for the cited gate theorems is

`propext`, `Classical.choice`, `Quot.sound`.

The checked sources contain no `sorry`, custom axiom declaration, or `native_decide`.

The table distinguishes the three evidence roles: ordinary proofs with independent formalization, kernel-checked finite certificates, and exact computer algebra outside Lean.

| Claim | Certificate and checker | Remaining trust |
|---|---|---|
| Defect, bounds, transfer, exact reconstruction, matching rigidity, bad-concurrence stability, and the two-unit design-nonexistence gap | `Gates.ArcsCompleteOutsideConicHuman` and `Gates.ArcsCompleteOutsideConicAdditions`; the named reconstruction and matching terminals in the two public trust documents | Lean kernel, Mathlib, and `propext`, `Classical.choice`, and `Quot.sound` |
| $q = 16$ exclusion | augmentation transitions, the line–triangle pattern certificate, $2630 + 3$ Lean leaf records, and `Q16QuadraticTransport`, imported by the separate certificate-package gate `Gates.ArcsCompleteOutsideConic` | The theorem trusts the Lean kernel and Mathlib. Exact Python arithmetic certifies the geometric compression independently but is not a premise of the Lean theorem; generator labels and deduplication are not trusted |
| Exact values at $q = 13, 17, 19$ | the exhaustive projective classifier and canonical JSON summaries; `Gates.ArcsCompleteOutsideConicAdditions` checks the three attaining witnesses and upper bounds | The upper bounds trust only the Lean kernel, Mathlib, and explicit finite-field reduction. The exhaustive lower classifications trust the C++ compiler and execution; the $q = 13$ lower bound has an independent frame enumeration, while the $q = 17, 19$ class reductions have no second implementation |
| Ten-point realization | `check_match10_rank_three.py` and its JSON certificate | Mathon's external two-class classification; Python integer arithmetic and Singular exact arithmetic; this claim is outside the Lean gate |
| Small-field witnesses | `ExampleChecks/Q5.lean` through `Q16.lean` | Lean kernel and the explicit finite-field implementations; no witness-search execution is trusted |

For $q = 13$ and $q = 17$, the small-order classifier fixes a projective four-frame, augments by every legal point, and retains the lexicographically least image over all ordered frames. Induction on the arc size therefore retains one representative of every projective class. At $q = 19$, it tests every canonical eight-arc and every legal extension of each one; deleting one point from an arbitrary nine-arc proves that this covers every nine-arc. The companion source is `small_odd_relative_conic_classifier.cpp`; the four canonical JSON summaries and `small_odd_relative_conic.sha256` record the parameters, counts, outputs, and replay hashes. A separate frame-normalized Python enumeration rechecks the $q = 13$ lower bound. No second implementation checks the $q = 17$ class reduction or the $q = 19$ extension reduction.

## F.2 Rank-three realization certificates

The ten-point bundle consists of `check_match10_rank_three.py`, its JSON certificate, and `check_match10_rank_three.sha256` in the paper directory. It is fixed at companion commit `ab3bcd6e`; the checksum file there records the full SHA-256 digests of the checker and JSON. The JSON fields `integral_lifts`, `algebraic_checks`, and `classical_char2_lex_basis` record respectively the rational unit relations and denominator primes, the saturated-ideal acceptance checks, and the characteristic-two triangular basis. The companion gives the exact replay command. Singular Gröbner bases and module lifts remain trusted executions. A second monomial order and a direct $\mathbb{F}_8$ incidence construction provide independent checks; the same record checks the regular hyperoval, the disjoint prescribed conic used in Corollary 4.6, and its nonvanishing gradient. Abstract two-class completeness remains the cited external theorem. The odd-characteristic computation saturates by $x_9(x_9-1)$, the product of the two necessary arc determinants displayed in the proof of Theorem A.3, and the checked rational module lifts have no denominator prime other than 2; hence their unit relations survive in every odd characteristic.

### F.3 The $q = 16$ augmentation and leaf certificates

For the $q = 16$ exclusion, the supplement contains the source generator `search_rhoc16.cpp`, its frozen report `search_rhoc16_output.txt`, and their SHA-256 digests. The nine-line report is only a generation summary, not the certificate: the proof objects are the emitted Lean transition and leaf modules consumed by the gate. Each certified transition carries an invertible matrix and checked source–target scalar equalities. The Lean checker separately verifies that these transitions cover every legal augmentation of the standard four-frame. At the leaves it checks either six independent quadratic evaluation conditions or an explicit linear combination forcing a containing quadratic to meet the arc. Thus the theorem trusts the Lean kernel and the stated classical inputs, but neither the generator's canonical labels nor an external claim of exhaustive execution.

The independent pattern checker recomputes every ordinary-uncovered locus from the frozen level-eight list and verifies the line–triangle pattern on exactly 2630 leaves. Its canonical JSON output records one six-point witness per leaf and the three exceptional kernels. This checker supplies the geometric compression used in the proof and checks the two displayed factorizations and the middle form's invertible standard-conic coordinate change on all $16^3$ coordinate vectors. The independently shaped Lean inverse matrices remain a kernel-checked verification of the same 2630+3 partition. The field-uniform normalized line–triangle argument is separately kernel-checked in `QuadraticLineTriangleObstruction`; the terminal theorem's determinant hypothesis is the coordinate form of noncollinearity for the three off-line points.

## G Finite-field witnesses

All coordinates below are relative to the standard conic

$$\mathcal{C}:\ XZ = Y^2.$$

For $q = 8$ and $q = 16$, an integer is the binary coefficient vector of the polynomial basis $1, \alpha, \ldots$, with $\alpha^3+\alpha+1 = 0$ and $\alpha^4+\alpha+1 = 0$, respectively. For $q = 9$, the integer $a_0+3a_1$ denotes $a_0 + a_1\alpha$, where $\alpha^2 + 1 = 0$. Coordinates for $q = 5, 11, 13, 17, 19$ are read modulo the prime. The checked witnesses are

| $q$ | $A$ |
|---|---|
| 5 | $\{(1,0,3),(2,1,2),(3,3,4),(1,4,4)\}$ |
| 8 | $\{(0,1,1),(0,1,2),(1,0,1),(1,0,2),(1,1,2),(1,1,6)\}$ |
| 9 | $\{(1,0,4),(1,0,5),(1,1,0),(1,1,2),(1,2,3),(1,2,4)\}$ |
| 11 | $\{(1,10,0),(1,9,1),(1,4,7),(1,8,5),(0,1,4),(1,1,7)\}$ |
| 13 | $\{(4,0,5),(3,12,7),(2,3,10),(9,2,9),$ $(7,1,0),(6,0,2),(8,12,11),(7,11,0)\}$ |
| 16 | $\{(1,5,14),(1,5,3),(1,9,2),(1,0,9),(1,13,13),$ $(1,6,10),(1,3,9),(0,1,11),(1,10,2)\}$ |
| 17 | $\{(0,11,0),(0,8,1),(3,5,15),(3,7,16),(3,3,0),$ $(3,0,1),(3,7,2),(3,12,7),(3,16,14)\}$ |
| 19 | $\{(0,1,8),(1,1,0),(1,3,14),(1,9,0),(1,9,12),$ $(1,11,11),(1,13,9),(1,16,1),(1,16,12),(1,18,7)\}$ |

The files `lean/RelativeConicArcs/ExampleChecks/Q5.lean` through `Q16.lean` check disjointness from $\mathcal{C}$, the arc condition, and relative coverage. The independent verifier `verify_relative_conic_arcs.py` regenerates the same coverage data; its digest and frozen output are recorded in the trust manifest. The public supplement gate `RelativeConicArcs/Gates/ArcsCompleteOutsideConicAdditions.lean` checks the three prime-field witnesses just added, including ordinary completeness of the $q = 19$ arc.

## AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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